

Vector Calculus

Mathematical Cookbook

MAST20029

Scalar Line Integrals

You will need:

1 - curve/contour C

1 - function $f(x,y)$

$$\int_C f(x,y) dS$$

Steps:

(1) Parameterise Curve

(2) Substitute your parameterisation of your curve into the function $f(x,y)$

(3) Expand out the factor dS to $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ (and substitute)

(4) Evaluate the integral

$$\int_C f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Example:

$$\int_C (x+y) dS$$

On the curve, $x = y^2$, from (0,0) to (1,1)

$$(1) \quad x = t \quad y = t^2 \quad 0 \leq t \leq 1$$

$$(2) \quad \int_C (x + \sqrt{y}) dS$$

$$\int_C (t + \sqrt{t^2}) dS$$

$$\int_C (2t) dS$$

$$(3) \quad \int_C (2t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 1 \quad \frac{dy}{dt} = 2t$$

$$(4) \quad \int_C (2t) \sqrt{(1)^2 + (2t)^2} dt \quad (\text{Evaluate via u-sub})$$

Vector Line Integrals

You will need:

- 1 - Curve/Contour C
- 1 - Vector Field $\mathbf{F}(x, y)$ - $\int_C \mathbf{F} \cdot d\mathbf{r}$

Steps:

- (1) Parameterise Curve
- (2) Substitute your parameterisation of your curve into your field $\mathbf{F}$
- (3) Expand out integral to

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} \cdot \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{pmatrix} dt$$
 (and substitute).
- (4) Evaluate the integral

Example:

*Example is in 2D, so no F_3 component.

Find: $\int_C \mathbf{F} \cdot d\mathbf{r}$

With: $\mathbf{F}(x, y) = (x + y, x^2 - y)$

On the curve traversing the upper half of the unit circle counterclockwise.

(1) $x = \cos t$ $y = \sin t$ $0 \leq t \leq \pi$

$\mathbf{r}(t) = (\cos t, \sin t)$

(2) $\mathbf{F}(x, y) = (x + y, x^2 - y)$

$\mathbf{F}(t) = (\cos t + \sin t, \cos^2 t - \sin t)$

(3)

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \begin{bmatrix} \cos t + \sin t \\ \cos^2 t - \sin t \end{bmatrix} \cdot \begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix} dt$$

$$\frac{dx}{dt} = -\sin t \quad \frac{dy}{dt} = \cos t$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \begin{bmatrix} \cos t + \sin t \\ \cos^2 t - \sin t \end{bmatrix} \cdot \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} dt$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^\pi (\cos t + \sin t)(-\sin t) + (\cos^2 t - \sin t)(\cos t) dt$$

Evaluate using trig-sub

Conservative Vector Fields

Testing whether a Vector Field is Conservative

You will need:

1 - Candidate vector field $\mathbf{F}$

Steps:

(1) Compute $\nabla \times \mathbf{F}$

(2) If $\nabla \times \mathbf{F} = 0 \rightarrow$ conservative.

Finding a Potential Function

You will need:

1 - Conservative Vector Field $\mathbf{F} = \langle F_1, F_2, F_3 \rangle$

Steps:

(1) Set:

$$\frac{\partial \phi}{\partial x} = F_1, \frac{\partial \phi}{\partial y} = F_2, \frac{\partial \phi}{\partial z} = F_3$$

(2) Integrate these three equations,

$$\text{To get: } \phi = \int F_1 dx + f(y, z), \phi = \int F_2 dy + g(x, z), \phi = \int F_3 dz + h(x, y)$$

(3) Fill in the missing terms between each of these equations to get ϕ

Evaluating a Line Integral on a Conservative Vector Field

You will need:

1 – Potential Function

2 – Endpoints you want to evaluate.

(1) Evaluate potential function at the two endpoints (and subtract):
 $\phi(x_2, y_2, z_2) - \phi(x_1, y_1, z_1)$

Important note: As conservative vector fields are defined/evaluated only by their endpoints, a line integral in a conservative vector field which is a closed loop is always 0.

Scalar Surface Integral

Explicit Version (Non-parametric):

You will need:

- (1) Function $z = g(x,y)$
- (1) Region of this surface you want to find the surface area of.

Steps:

- (1) Write out the surface integral expression:

$$\iint_S g(x,y) dS$$

- (2) Expand out dS term to:

$$\iint_S g(x,y) \sqrt{\left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2 + 1} dx dy$$

- (3) Find bounds of region and integrate as a double integral.

Vector Surface Integral

Explicit:

You will need:

(1) function $z = g(x,y)$

(1) vector field $\mathbf{F} = \langle F_1, F_2, F_3 \rangle$

Steps:

(1) Write out the surface integral expression:

$$\iint_S \mathbf{F} \cdot d\mathbf{S}$$

(2) Expand out the surface integral expression, and evaluate the dot product:

(3) Find bounds of surface and evaluate remaining integral as a regular double integral.

Stokes' Theorem

(via situation where you compute curl*, explicit (non-parametric version))

You will need:

(1) A function $z = f(x,y)$.

(1) Vector Field $\mathbf{F}$

Steps:

(1) Evaluate Curl ($\nabla \times \mathbf{F}$) of vector field.

(2) Find bounds of the region of 3D space you are evaluating

(3) Expand dS to:

$$\text{With: } \hat{n} = \begin{bmatrix} -f_x \\ -f_y \\ 1 \end{bmatrix}$$

$$\iint_S (\nabla \times \mathbf{F}) dS = \iint_S ((\nabla \times \mathbf{F}) \cdot \hat{n}) dy dx$$

For an upwards facing normal.

(4) Evaluate dot product and integrate (as a regular double integral):

$$\iint_S ((\nabla \times \mathbf{F}) \cdot \begin{bmatrix} -f_x \\ -f_y \\ 1 \end{bmatrix}) dy dx$$

*Important note: There are various ways you may need to invoke Stokes' Theorem. This is the most common use case, however! (usually for Stokes' you are going to either compute a double integral (with curl) or a line integral directly and then invoke the theorem!)

Divergence/Gauss's Theorem

(via situation where you compute divergence*)

You will need:

(1) Region of 3D space.

(1) Vector Field $\mathbf{F}$

Steps:

(1) Evaluate Divergence ($\nabla \cdot \mathbf{F}$) of vector field.

(2) Find bounds of the region of 3D space you are evaluating

(3) Evaluate (either directly or through changing your coordinates):

$$\iiint_V \nabla \cdot \mathbf{F} dV$$

Important note: There are various ways you may need to invoke Divergence Theorem. This is the most common use case, however! (usually for Divergence Theorem you are going to either compute a triple integral (with divergence) or a surface integral directly and then invoke the theorem!)